The normal distribution

In this lab, you'll investigate the probability distribution that is most central to statistics: the normal distribution. If you are confident that your data are nearly normal, that opens the door to many powerful statistical methods. Here we'll use the graphical tools of R to assess the normality of our data and also learn how to generate random numbers from a normal distribution.

Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages as well as the **openintro** package.

Let's load the packages.

```
library(tidyverse)
library(openintro)
```

The data

This week you'll be working with fast food data. This data set contains data on 515 menu items from some of the most popular fast food restaurants worldwide. Let's take a quick peek at the first few rows of the data.

Either you can use `glimpse` like before, or `head` to do this.

```
library(tidyverse)
library(openintro)
data("fastfood", package='openintro')
head(fastfood)
```

```
## # A tibble: 6 x 17
##   restaurant item       calories cal_fat total_fat sat_fat trans_fat cholesterol
##   <chr>      <chr>         <dbl>   <dbl>     <dbl>   <dbl>     <dbl>       <dbl>
## 1 Mcdonalds  Artisan G~      380      60         7       2       0            95
## 2 Mcdonalds  Single Ba~      840     410        45      17       1.5         130
## 3 Mcdonalds  Double Ba~     1130     600        67      27       3           220
## 4 Mcdonalds  Grilled B~      750     280        31      10       0.5         155
## 5 Mcdonalds  Crispy Ba~      920     410        45      12       0.5         120
## 6 Mcdonalds  Big Mac         540     250        28      10       1            80
## # i 9 more variables: sodium <dbl>, total_carb <dbl>, fiber <dbl>, sugar <dbl>,
## #   protein <dbl>, vit_a <dbl>, vit_c <dbl>, calcium <dbl>, salad <chr>
```

You'll see that for every observation there are 17 measurements, many of which are nutritional facts.

You'll be focusing on just three columns to get started: restaurant, calories, calories from fat.

Let's first focus on just products from McDonalds and Dairy Queen.

```r
mcdonalds <- fastfood %>%
  filter(restaurant == "Mcdonalds")
dairy_queen <- fastfood %>%
  filter(restaurant == "Dairy Queen")
```

1. Make a plot (or plots) to visualize the distributions of the amount of calories from fat of the options from these two restaurants. How do their centers, shapes, and spreads compare?

Insert your answer here

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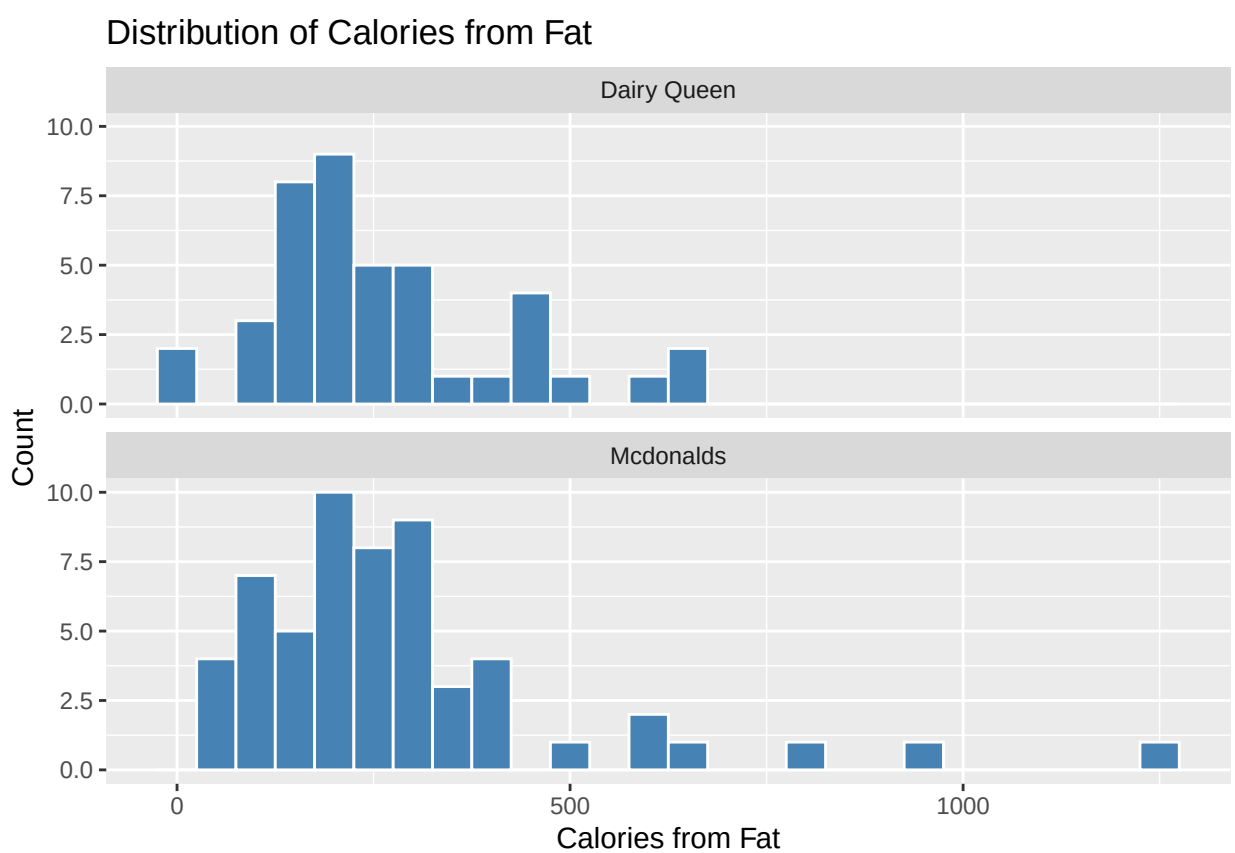
```r
# Side-by-side histograms
ggplot(data = fastfood %>% filter(restaurant %in% c("Mcdonalds", "Dairy Queen")),
       aes(x = cal_fat)) +
  geom_histogram(binwidth = 50, fill = "steelblue", color = "white") +
  facet_wrap(~ restaurant, ncol = 1) +
  labs(title = "Distribution of Calories from Fat",
       x = "Calories from Fat", y = "Count")
```




```r
# Summary statistics
mcdonalds %>% summarise(mean = mean(cal_fat), median = median(cal_fat), sd = sd(cal_fat), IQR = IQR(cal
```

```
## # A tibble: 1 x 4
```

```
##    mean median    sd   IQR
##   <dbl>  <dbl> <dbl> <dbl>
## 1  286.    240  221.   160
```

```
dairy_queen %>% summarise(mean = mean(cal_fat), median = median(cal_fat), sd = sd(cal_fat), IQR = IQR(ca
```

```
## # A tibble: 1 x 4
##    mean median    sd   IQR
##   <dbl>  <dbl> <dbl> <dbl>
## 1  260.    220  156.   150
```

Both distributions are unimodal and right-skewed. McDonald's has a higher center (mean ~ 286, median ~ 240) compared to Dairy Queen (mean ~ 260, median ~ 220). McDonald's also has a larger spread (sd ~ 221) compared to Dairy Queen (sd ~ 156). Both distributions have a longer right tail, indicating some menu items with very high calories from fat.

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The normal distribution

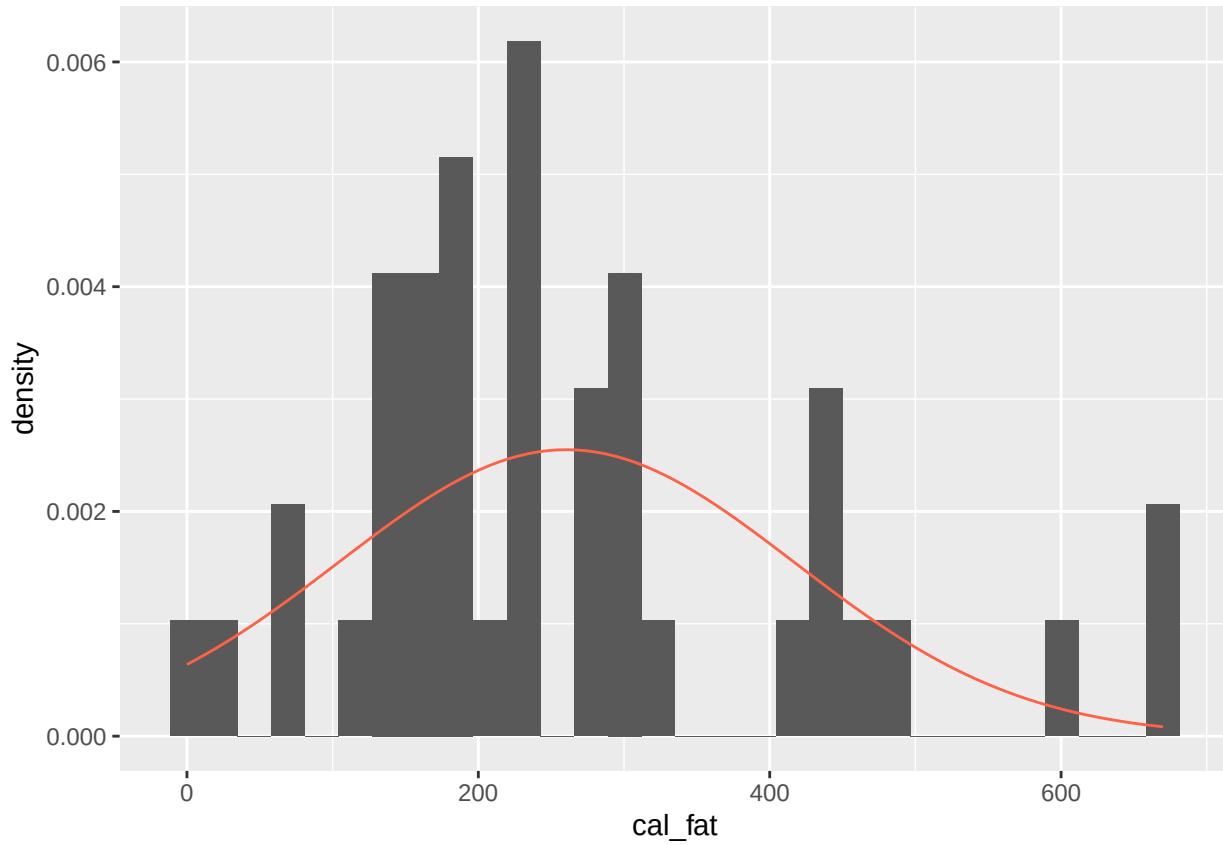
In your description of the distributions, did you use words like *bell-shaped*or *normal*? It's tempting to say so when faced with a unimodal symmetric distribution.

To see how accurate that description is, you can plot a normal distribution curve on top of a histogram to see how closely the data follow a normal distribution. This normal curve should have the same mean and standard deviation as the data. You'll be focusing on calories from fat from Dairy Queen products, so let's store them as a separate object and then calculate some statistics that will be referenced later.

```
dqmean <- mean(dairy_queen$cal_fat)
dqsd   <- sd(dairy_queen$cal_fat)
```

Next, you make a density histogram to use as the backdrop and use the `lines` function to overlay a normal probability curve. The difference between a frequency histogram and a density histogram is that while in a frequency histogram the *heights* of the bars add up to the total number of observations, in a density histogram the *areas* of the bars add up to 1. The area of each bar can be calculated as simply the height *times* the width of the bar. Using a density histogram allows us to properly overlay a normal distribution curve over the histogram since the curve is a normal probability density function that also has area under the curve of 1. Frequency and density histograms both display the same exact shape; they only differ in their y-axis. You can verify this by comparing the frequency histogram you constructed earlier and the density histogram created by the commands below.

```
ggplot(data = dairy_queen, aes(x = cal_fat)) +
        geom_blank() +
        geom_histogram(aes(y = ..density..)) +
        stat_function(fun = dnorm, args = c(mean = dqmean, sd = dqsd), col = "tomato")
```



After initializing a blank plot with `geom_blank()`, the `ggplot2` package (within the `tidyverse`) allows us to add additional layers. The first layer is a density histogram. The second layer is a statistical function – the density of the normal curve, `dnorm`. We specify that we want the curve to have the same mean and standard deviation as the column of fat calories. The argument `col` simply sets the color for the line to be drawn. If we left it out, the line would be drawn in black.

2. Based on the this plot, does it appear that the data follow a nearly normal distribution?

Insert your answer here

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The histogram of Dairy Queen's calories from fat is roughly unimodal and somewhat bell-shaped, so the data do appear to follow a nearly normal distribution.

However, the distribution shows some right skew — the right tail extends further than the left, and there are a few high-calorie items that pull the distribution away from perfect normality.

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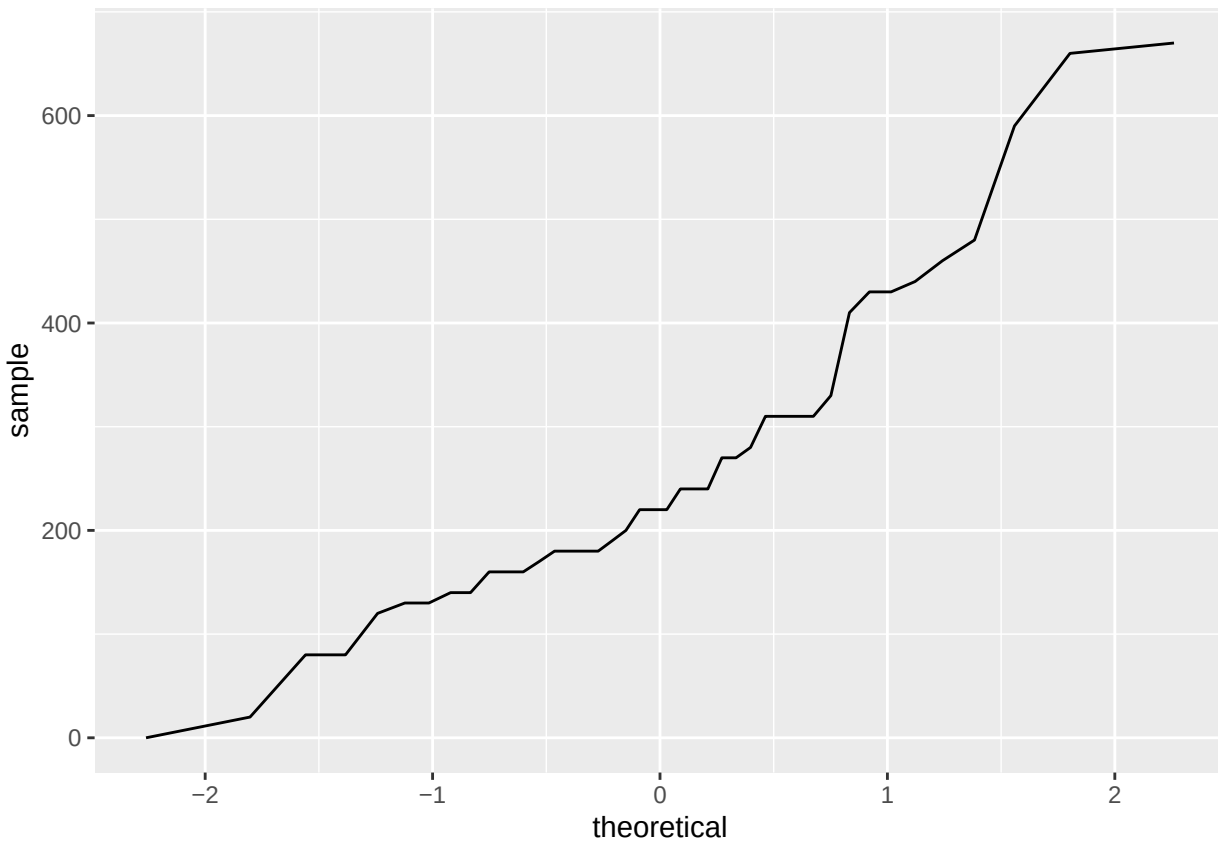
Evaluating the normal distribution

Eyeballing the shape of the histogram is one way to determine if the data appear to be nearly normally distributed, but it can be frustrating to decide just how close the histogram is to the curve. An alternative approach involves constructing a normal probability plot, also called a normal Q-Q plot for "quantile-quantile".

```r
ggplot(data = dairy_queen, aes(sample = cal_fat)) +
  geom_line(stat = "qq")
```



This time, you can use the `geom_line()` layer, while specifying that you will be creating a Q-Q plot with the `stat` argument. It's important to note that here, instead of using `x` instead `aes()`, you need to use `sample`.

The x-axis values correspond to the quantiles of a theoretically normal curve with mean 0 and standard deviation 1 (i.e., the standard normal distribution). The y-axis values correspond to the quantiles of the original unstandardized sample data. However, even if we were to standardize the sample data values, the Q-Q plot would look identical. A data set that is nearly normal will result in a probability plot where the points closely follow a diagonal line. Any deviations from normality leads to deviations of these points from that line.

The plot for Dairy Queen's calories from fat shows points that tend to follow the line but with some errant points towards the upper tail. You're left with the same problem that we encountered with the histogram above: how close is close enough?

A useful way to address this question is to rephrase it as: what do probability plots look like for data that I *know* came from a normal distribution? We can answer this by simulating data from a normal distribution using `rnorm`.

```r
sim_norm <- rnorm(n = nrow(dairy_queen), mean = dqmean, sd = dqsd)
```

The first argument indicates how many numbers you'd like to generate, which we specify to be the same number of menu items in the `dairy_queen` data set using the `nrow()` function. The last two arguments determine the mean and standard deviation of the normal distribution from which the simulated sample will

be generated. You can take a look at the shape of our simulated data set, `sim_norm`, as well as its normal probability plot.

3. Make a normal probability plot of `sim_norm`. Do all of the points fall on the line? How does this plot compare to the probability plot for the real data? (Since `sim_norm` is not a data frame, it can be put directly into the `sample` argument and the `data` argument can be dropped.)

Insert your answer here

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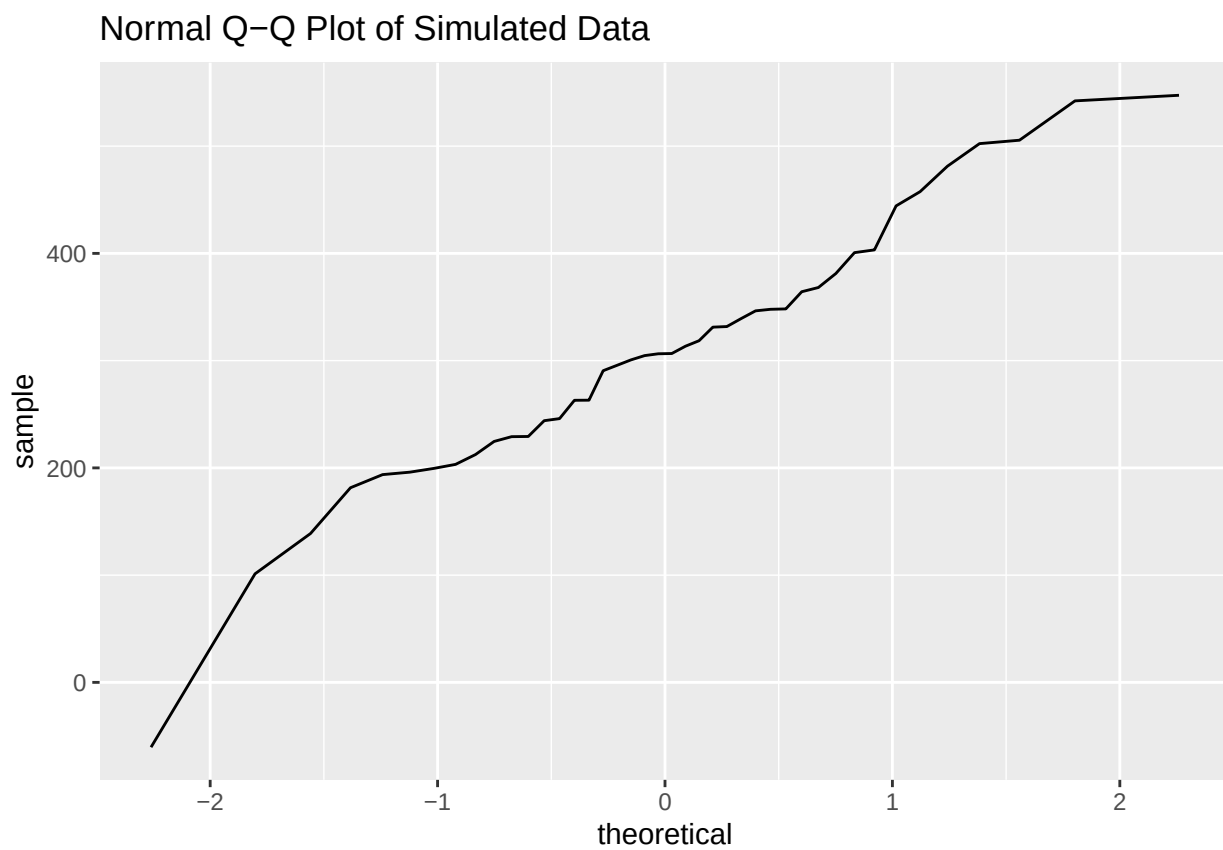
```
ggplot(data = NULL, aes(sample = sim_norm)) +
  geom_line(stat = "qq") +
  labs(title = "Normal Q-Q Plot of Simulated Data")
```



Not all of the points fall exactly on the line, even though the data were generated from a perfect normal distribution.

The points follow the diagonal line much more closely overall than the real Dairy Queen data. The real data showed more noticeable deviations in the upper tail, while the simulated data's departures from the line are smaller and more random. -> Even truly normal data won't produce a perfectly straight Q-Q plot.

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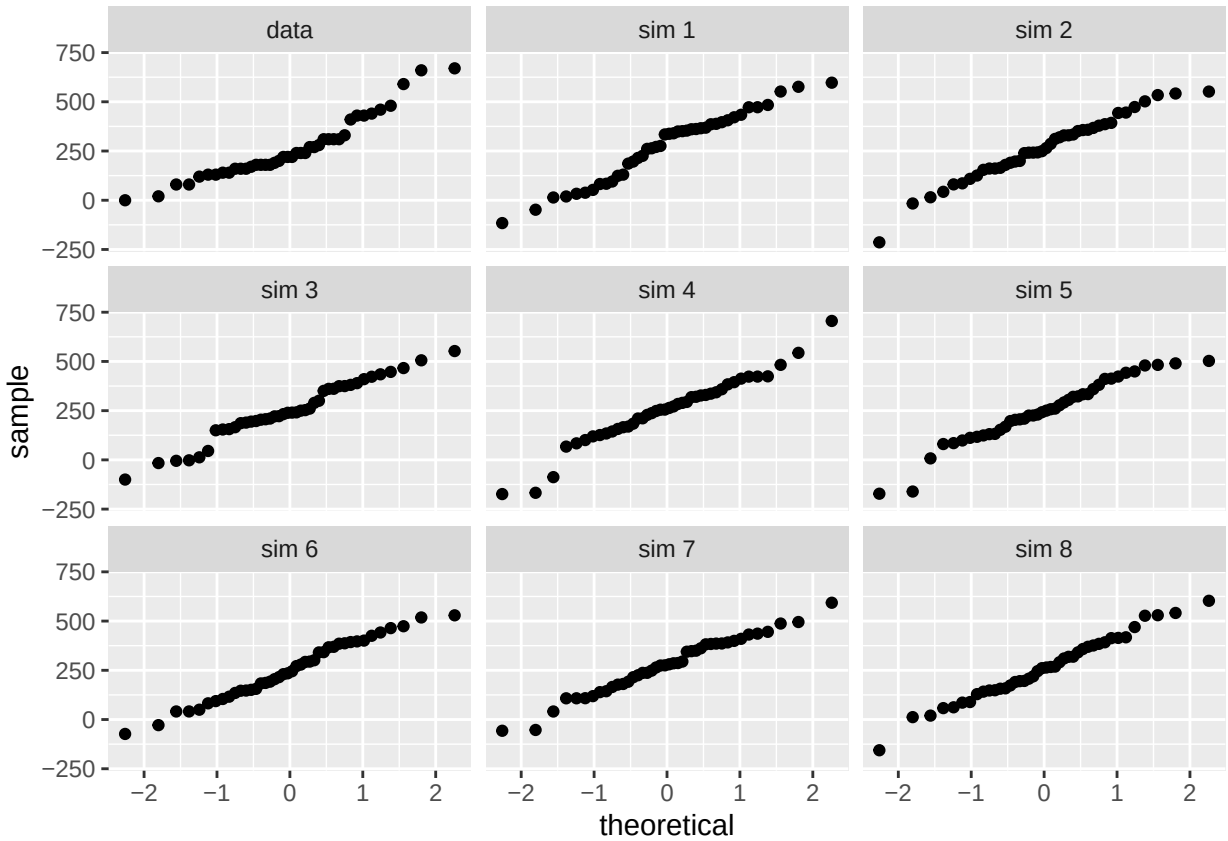
Even better than comparing the original plot to a single plot generated from a normal distribution is to compare it to many more plots using the following function. It shows the Q-Q plot corresponding to the original data in the top left corner, and the Q-Q plots of 8 different simulated normal data. It may be helpful to click the zoom button in the plot window.

```r
qqnormsim(sample = cal_fat, data = dairy_queen)
```



4. Does the normal probability plot for the calories from fat look similar to the plots created for the simulated data? That is, do the plots provide evidence that the calories are nearly normal?

Insert your answer here

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Yes, the normal probability plot for Dairy Queen's calories from fat looks reasonably similar to the simulated Q-Q plots.

The points in the real data generally follow the diagonal line in the central portion, similar to what we see in the simulated plots. Overall, the plots provide evidence that the calories from fat are approximately normally distributed, though not perfectly so.

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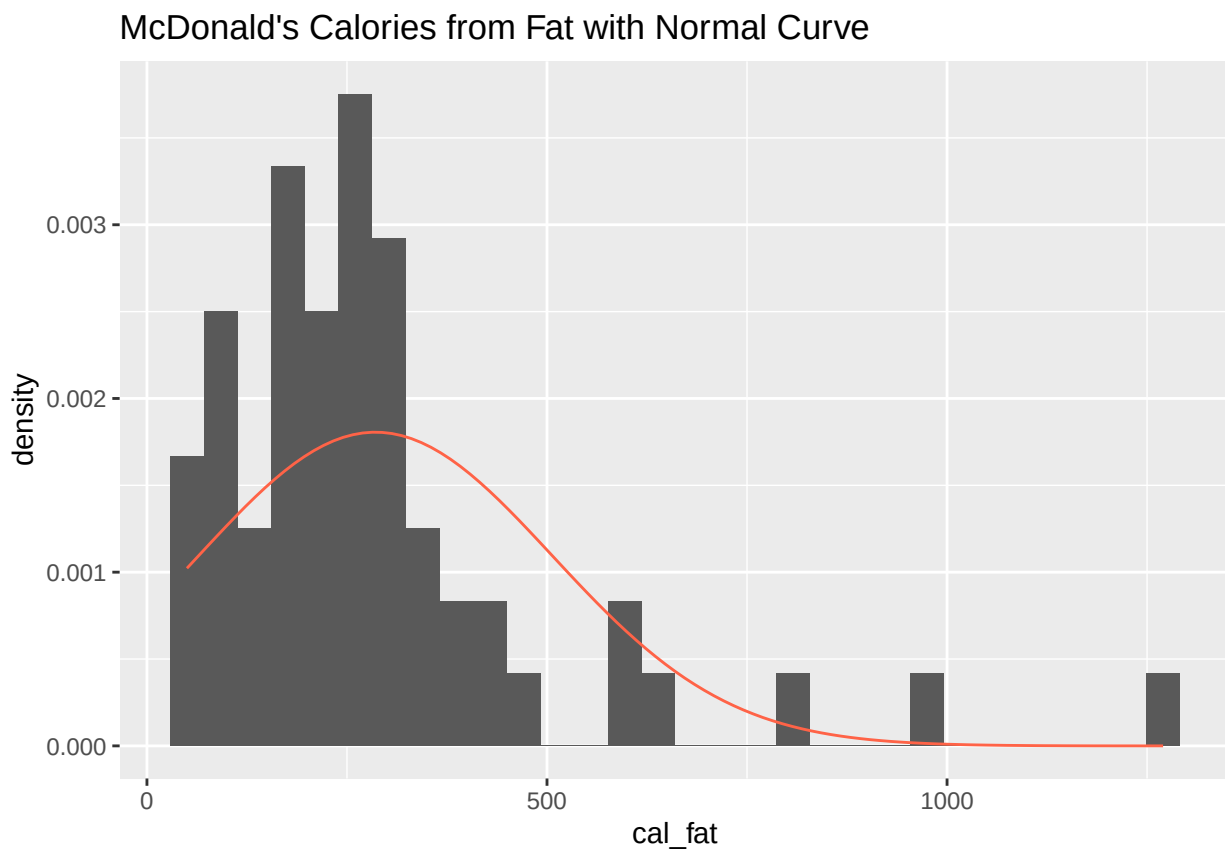
5. Using the same technique, determine whether or not the calories from McDonald's menu appear to come from a normal distribution.

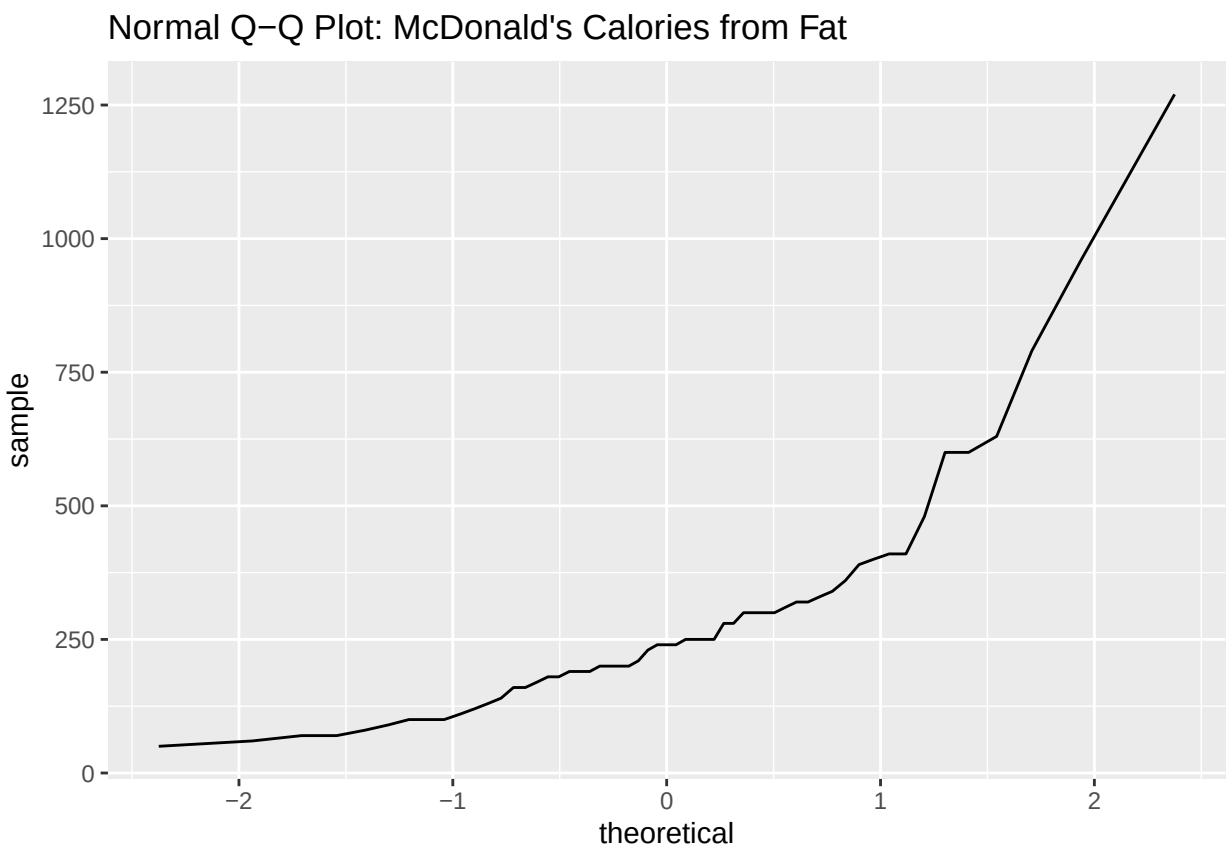
Insert your answer here

==========

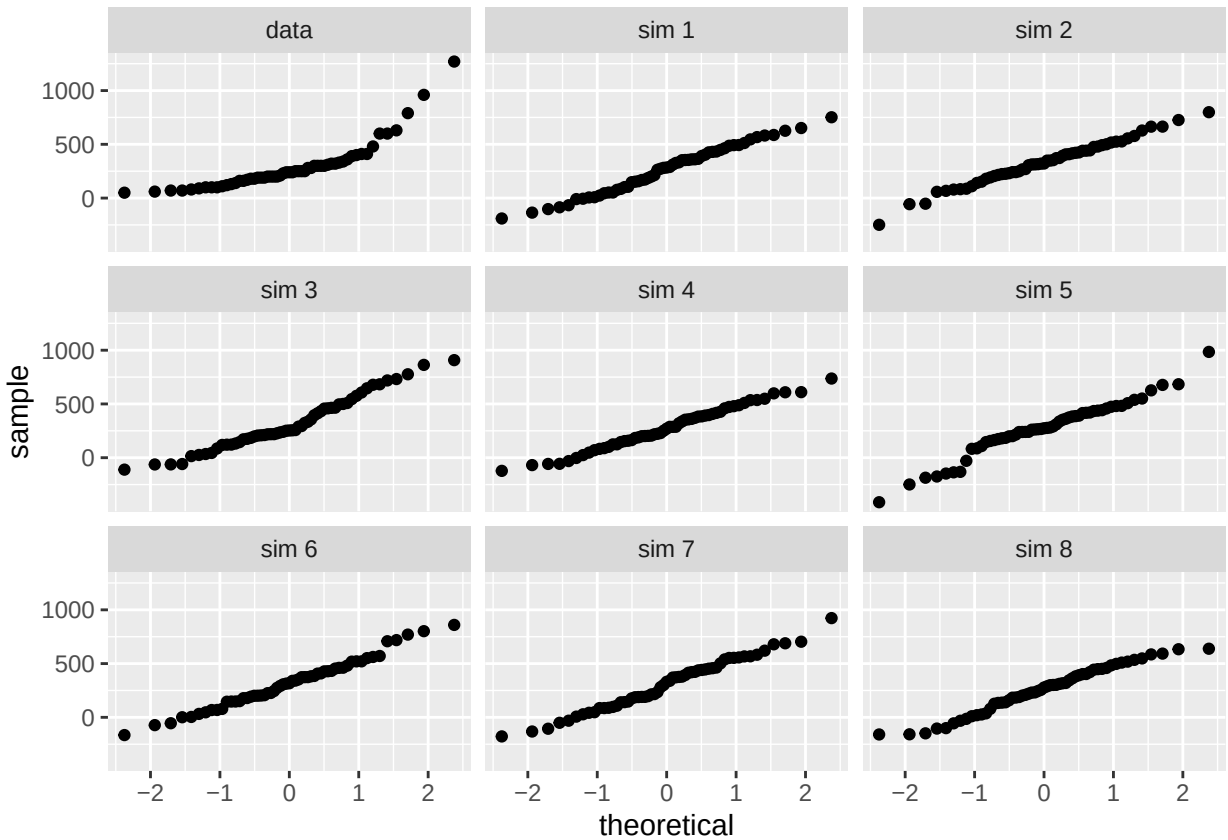
```r
# Histogram with normal curve overlay
mcmean <- mean(mcdonalds$cal_fat)
mcsd <- sd(mcdonalds$cal_fat)

ggplot(data = mcdonalds, aes(x = cal_fat)) +
  geom_blank() +
  geom_histogram(aes(y = ..density..)) +
  stat_function(fun = dnorm, args = c(mean = mcmean, sd = mcsd), col = "tomato") +
  labs(title = "McDonald's Calories from Fat with Normal Curve")
```




```r
# Q-Q plot
ggplot(data = mcdonalds, aes(sample = cal_fat)) +
  geom_line(stat = "qq") +
  labs(title = "Normal Q-Q Plot: McDonald's Calories from Fat")
```




```r
# qqnormsim comparison
qqnormsim(sample = cal_fat, data = mcdonalds)
```



The histogram for McDonald's calories from fat shows a right-skewed distribution, the normal curve does not fit as well as it did for Dairy Queen, with the peak shifted to the left.

The Q-Q plot confirms this: the points curve upward away from the line in the upper tail, indicating right skew.

Compared to the simulated normal Q-Q plots, the McDonald's data shows more systematic deviation from the line. Therefore, the calories from fat for McDonald's do not appear to follow a normal distribution as closely as Dairy Queen's data; the distribution is notably right-skewed.

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Normal probabilities

Okay, so now you have a slew of tools to judge whether or not a variable is normally distributed. Why should you care?

It turns out that statisticians know a lot about the normal distribution. Once you decide that a random variable is approximately normal, you can answer all sorts of questions about that variable related to probability. Take, for example, the question of, "What is the probability that a randomly chosen Dairy Queen product has more than 600 calories from fat?"

If we assume that the calories from fat from Dairy Queen's menu are normally distributed (a very close approximation is also okay), we can find this probability by calculating a Z score and consulting a Z table (also called a normal probability table). In R, this is done in one step with the function `pnorm()`.

```
1 - pnorm(q = 600, mean = dqmean, sd = dqsd)
```

```
## [1] 0.01501523
```

Note that the function `pnorm()` gives the area under the normal curve below a given value, `q`, with a given mean and standard deviation. Since we're interested in the probability that a Dairy Queen item has more than 600 calories from fat, we have to take one minus that probability.

Assuming a normal distribution has allowed us to calculate a theoretical probability. If we want to calculate the probability empirically, we simply need to determine how many observations fall above 600 then divide this number by the total sample size.

```r
dairy_queen %>%
  filter(cal_fat > 600) %>%
  summarise(percent = n() / nrow(dairy_queen))
```

```
## # A tibble: 1 x 1
##   percent
##     <dbl>
## 1  0.0476
```

Although the probabilities are not exactly the same, they are reasonably close. The closer that your distribution is to being normal, the more accurate the theoretical probabilities will be.

6. Write out two probability questions that you would like to answer about any of the restaurants in this dataset. Calculate those probabilities using both the theoretical normal distribution as well as the empirical distribution (four probabilities in all). Which one had a closer agreement between the two methods?

Insert your answer here

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Question 1:

What is the probability that a randomly chosen Dairy Queen item has fewer than 150 calories from fat?

```r
# Theoretical probability
pnorm(q = 150, mean = dqmean, sd = dqsd)
```

```
## [1] 0.2400986
```

```r
# Empirical probability
dairy_queen %>%
  filter(cal_fat < 150) %>%
  summarise(percent = n() / nrow(dairy_queen))
```

```
## # A tibble: 1 x 1
##   percent
##     <dbl>
## 1   0.214
```

Question 2:

What is the probability that a randomly chosen McDonald's item has more than 400 calories from fat?

```r
# Theoretical probability
1 - pnorm(q = 400, mean = mcmean, sd = mcsd)
```

```
## [1] 0.3022921
```

```r
# Empirical probability
mcdonalds %>%
  filter(cal_fat > 400) %>%
  summarise(percent = n() / nrow(mcdonalds))
```

```
## # A tibble: 1 x 1
##   percent
##     <dbl>
## 1   0.158
```

Question 1 (Dairy Queen, fewer than 150 cal from fat) tends to have closer agreement between the theoretical and empirical probabilities, because Dairy Queen's calorie-from-fat distribution is closer to normal than McDonald's.

When the data follow the normal distribution more closely, the theoretical probability calculated using `pnorm` will better match the empirical proportion from the data. McDonald's right-skewed distribution means the normal model is less accurate, so the two methods show more disagreement for Question 2.

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More Practice

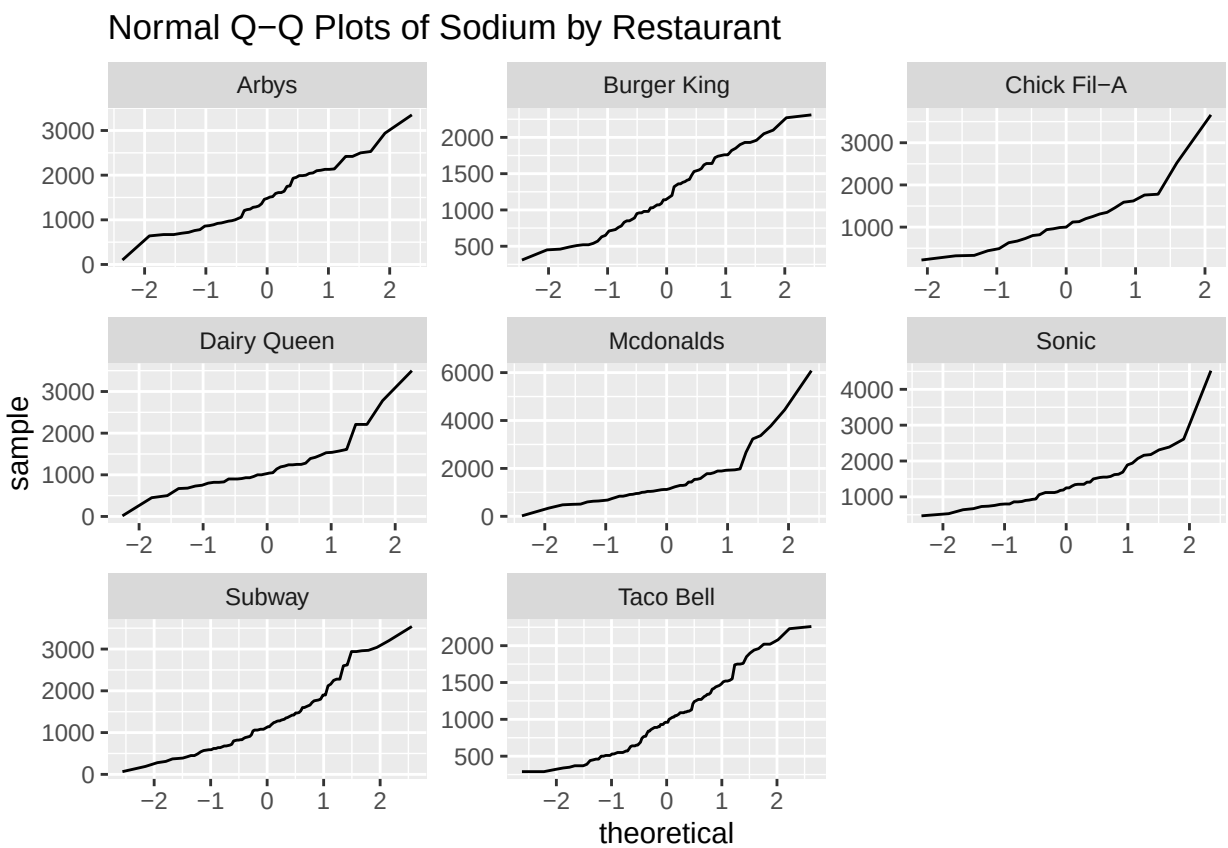
7. Now let's consider some of the other variables in the dataset. Out of all the different restaurants, which ones' distribution is the closest to normal for sodium?

Insert your answer here

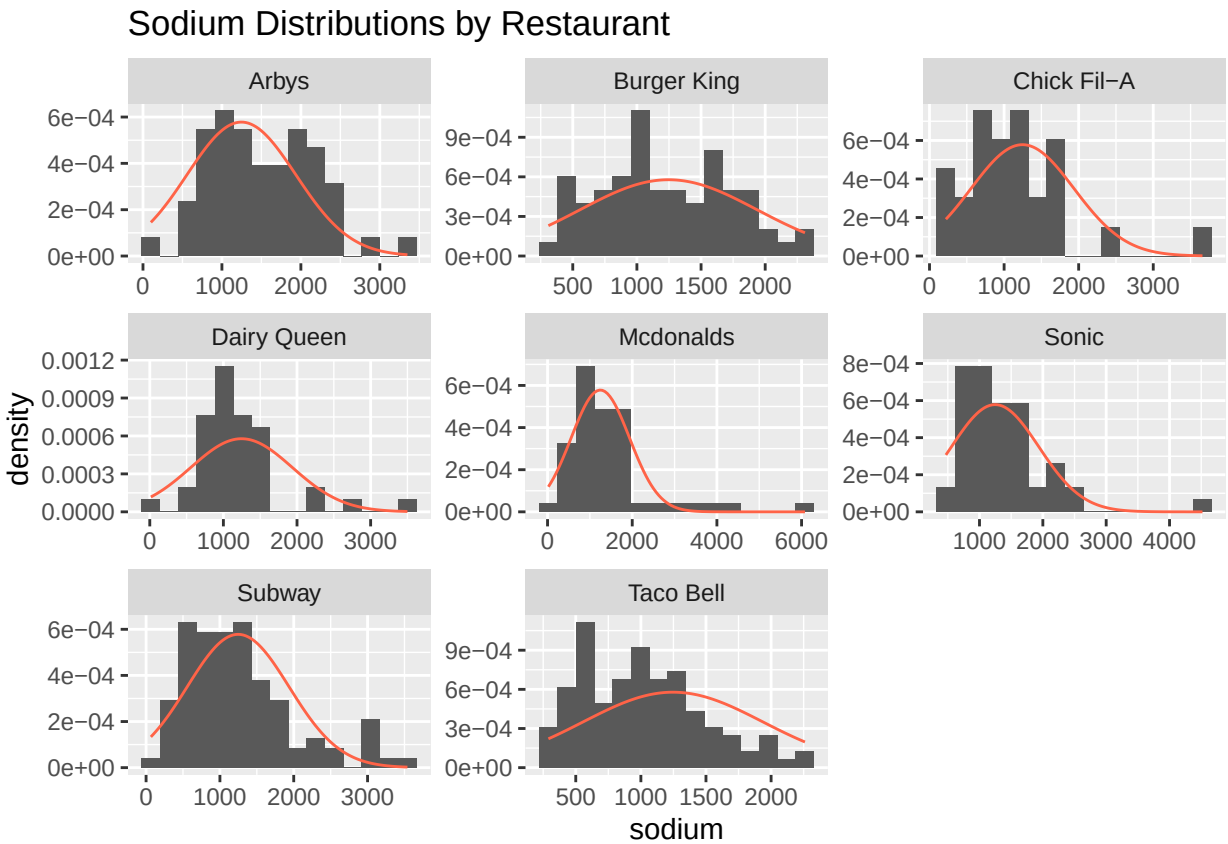
==========

```r
# Get all unique restaurants
restaurants <- unique(fastfood$restaurant)

# Create Q-Q plots for sodium by restaurant
ggplot(data = fastfood, aes(sample = sodium)) +
  geom_line(stat = "qq") +
  facet_wrap(~ restaurant, scales = "free") +
  labs(title = "Normal Q-Q Plots of Sodium by Restaurant")
```




```r
# Also check histograms
ggplot(data = fastfood, aes(x = sodium)) +
  geom_histogram(aes(y = ..density..), bins = 15) +
  facet_wrap(~ restaurant, scales = "free") +
  stat_function(fun = dnorm,
                args = list(mean = mean(fastfood$sodium, na.rm = TRUE),
                            sd = sd(fastfood$sodium, na.rm = TRUE)),
                col = "tomato") +
  labs(title = "Sodium Distributions by Restaurant")
```



Based on the Q-Q plots, **Burger King** and **Taco Bell** tend to have sodium distributions that are closest to normal.

Their Q-Q plots show points that follow the diagonal line most closely with relatively minor deviations in the tails.

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8. Note that some of the normal probability plots for sodium distributions seem to have a stepwise pattern. why do you think this might be the case?

Insert your answer here

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Because sodium values in fast food items tend to come in discrete, rounded amounts rather than continuous values.

Restaurants often have menu items with similar sodium levels (e.g., many items clustered at 500 mg, 1000 mg, 1500 mg, etc.) because items share common ingredients and preparation methods.

When multiple observations take the same or very similar values, they appear as flat "steps" (horizontal segments) on the Q-Q plot since the y-axis values stay constant while the theoretical quantiles on the x-axis continue to increase.

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9. As you can see, normal probability plots can be used both to assess normality and visualize skewness. Make a normal probability plot for the total carbohydrates from a restaurant of your choice. Based on

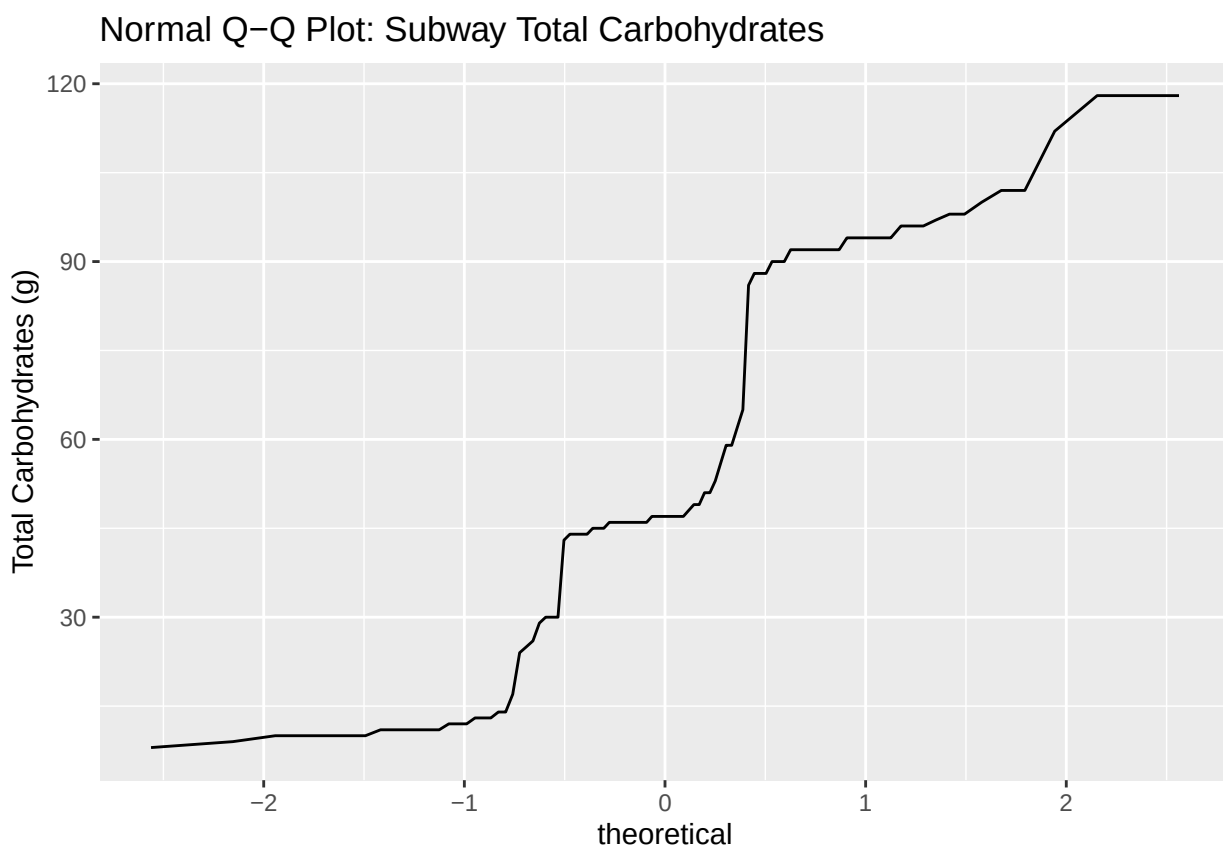
this normal probability plot, is this variable left skewed, symmetric, or right skewed? Use a histogram to confirm your findings.

Insert your answer here

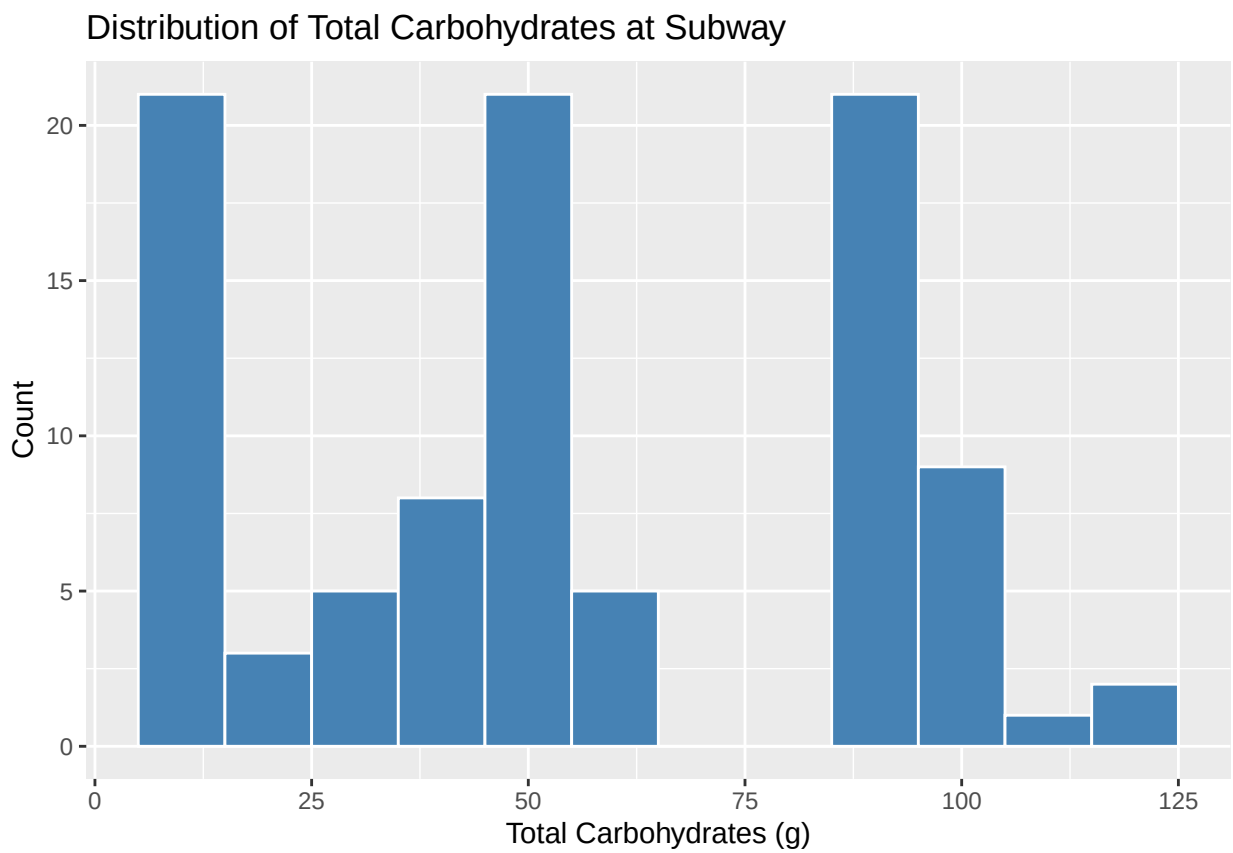
==========

```
subway <- fastfood %>% filter(restaurant == "Subway")

# Normal probability (Q-Q) plot for total carbs
ggplot(data = subway, aes(sample = total_carb)) +
  geom_line(stat = "qq") +
  labs(title = "Normal Q-Q Plot: Subway Total Carbohydrates",
       y = "Total Carbohydrates (g)")
```




```
# Histogram to confirm
ggplot(data = subway, aes(x = total_carb)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Total Carbohydrates at Subway",
       x = "Total Carbohydrates (g)", y = "Count")
```



Right skew: the Q-Q plot for Subway's total carbohydrates shows the points curving upward above the line at the upper end and staying close to or below the line at the lower end.

The histogram confirms this finding: the distribution has a peak on the left side with a long tail stretching to the right, indicating that most Subway items have moderate carbohydrate levels but a few items have much higher carb counts.

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